

ICE HOCKEY

DATA PREPARATION

Knowledge Discovery & Data Mining · TU Graz

• **Group 100** • • • •

approach to unknown data

Most of the given features are well known, especially for someone who played a similar sport before.

However, some features were fully or partly unknown, but either `feature_description.txt`, researching or looking at the data distribution was enough to get an understanding. Examples:

- time span of salary (month/year), but since values range between [124k – 18M], it is assumed yearly
- `num_of_shots`: count of attempts that would have entered the net if not stopped by the goaltender (researched)
- `time_on_ice`: values 3-30 are presumably minutes, when a regular match lasts 60 minutes (researched)

This gave enough information to do data preprocessing and cleaning, while still unknown details do not matter for model training later, like: currency of salary (€/ \$) does not matter since its similar, exact criteria of `high_danger_shots`, ...

For data analysis we used:

- **pandas** (columns, datatypes, count of outliers, ...)
- **Data Wrangler** (vscode: displays dataframe, sort, filter)
- **Orange** (Histogram, Scatter Plot, ...)

preprocessing 1/2

A large portion of our preprocessing is bringing the data into a consistent format with correct datatypes:

- m, male, f, female (str) → male, female (category)
- capitalizing city/country, removing dot, e.g. [usa, U.S.A.] -> [Usa, Usa]
- correcting typos: Norwayy, Kazn, Tmpere
- converting height {m, cm} → {cm}
- removing “%”, “cm”, “m”, “kg” for str → number
- “nan” (str) → actual NaN/null
- first 2 rows of df_medical_information contain headers
- data type float64 → Int64, category, bool
- date conversion with pd.to_datetime(..., format='mixed')
- rounding javascript floats: 3.18000000000002 → 3.18

preprocessing 2/2

df_identy_card_0 is in roman/latin

- headers are in latin, e.g. “aetas_annorum” → “age”
- medical_id is in roman numerals, e.g. “MMMCMLXV” → 3965
- first_name had suffix “us”
- last_name had suffix “ius”

df_identity_card_0 and **df_identity_card_1** contain exclusive entries, so no overlap, and their combination (concat) are exactly the 10.000 desired unique international_id entries. So after the header renaming of above, they can be concat'ed to one dataframe **df_identity_cards**

data relationship 1/2

Generally, **international_id** is used to identify entities.

Exception 1: df_medical_information only has **medical_id**, but df_identity_cards has both international_id and medical_id, so we can link them.

Exception 2: df_moms_notes does not have any of the two id columns. A combination of the keys **['first_name', 'last_name', 'age', 'birth_city']** was used to link data entries. Any combination of fewer keys has many duplicates. Our **keyset** only has one duplicate (Lillian). Our cold hearted solution is to just remove one entry, since its not that significant.

For df_moms_notes, a left join was chosen because moms_notes is negletable if missing or not assignable. However, no mom_notes were missing anyways.

data relationship 2/2

merge statistic:

left_df	right_df	join_keys	left rows	right rows	left unique	right unique	matching keys	left-only keys	right-only keys
contracts	hockey_scouting	international_id	10000	10000	10000	10000	10000	0	0
contracts + scouting	identity_cards	international_id	10000	1000	10000	10000	10000	0	0
previous merge	medical_information	medical_id	10000	① 9963	10000	9963	9963	① 37	0
previous merge	performance	international_id	10000	10000	10000	10000	10000	0	0
previous merge	moms_notes	first_name, last_name, birth_city, age	10000	② 9962	③ 9963	9962	8957	④ 1	0

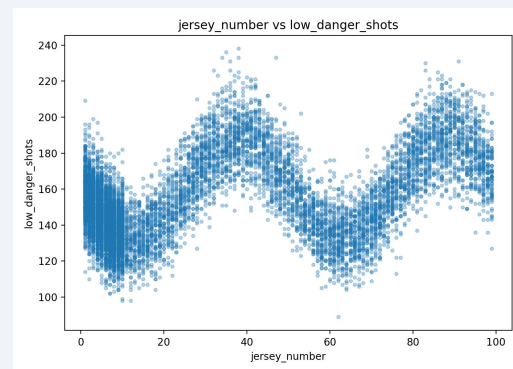
- ① 37 entries don't have medical_information. The rows exist (originally), but they only have meaningful med_id and signature.
- ② Additional to the 37 empty rows, we had to remove one duplicate for consistency.
- ③ These are again the 37 entries with almost all null values, which includes the 4 keys.
- ④ This 1 key-set that is only in df_total are the 37 null rows which were already removed in df_moms_notes.

These are the top non-obvious **correlations**. We measured Pearson, Spearman and others.

Feature 1	Feature 2	Pearson	Spearman
goals	shooting_percentage_calculated	0.7806724508	0.976641657
power_play_goals	shooting_percentage_calculated	0.6873173613	0.8365672204
passes_completed	pass_completion_rate	0.7459179587	0.7980837824
height	body_fat_percentage	-0.7068886786	-0.6919223074
assists	games_missed_due_to_injury	0.6884917764	0.6960161917
time_on_ice	time_between_penalties	0.6246230248	0.6859392293
shoe_size	body_fat_percentage	-0.6501597607	-0.6338771962

more thorough and complete data is generated by our code.

special correlation example found



data cleaning/imputation (missing values, outliers, ...) 1/2

37 ghost players: 37 entries have all NaN values, except captain (always True) and won_championship (always True)

Additional Missing Values Results

medical_information field: 85% no injuries (8519 out of 10000),
has_medical_note convert to binary field

fitness_level: 467 missing (37 ghosts + 322 actual missing +
108 NaN “strings” converted to actual NaNs)

physician_signature: All records were (joke data) removed
entirely

moms_notes: Removal of 3 out of 13 total columns
(favourite_child_info, favourite_tv_show, favourite_food)
because they contained either no or the same value for all.

Contradictions

- shooting_percentage != 100 * goals / num_of_shots ($\pm 1\%$): 9511
- high + medium + low danger shots > num_of_shots: 8153
- high_danger_shots > num_of_shots: 58
- medium_danger_shots > num_of_shots: 59
- low_danger_shots > num_of_shots: 60
- power_play_goals > goals: 8630
- power_play_time > time_on_ice: 542
- puck_recoveries > puck_touches: 2650
- penalty_kill_time > time_on_ice: 841
- power_play_time + penalty_kill_time > time_on_ice: 3538
- puck_possession_time > time_on_ice: 3922
- penalties_taken > 0 and penalty_minutes == 0: 151
- penalty_minutes > 0 and penalties_taken == 0: 5762
- years_in_usa > age: 1

TOTAL HARD CONTRADICTIONS: 43878

NUMBER OF CONTRADICTING CONDITIONS: 14

since shooting_percentage is so inaccurate, we created a correct
shooting_percentage_calculated which is $(100 * \text{goals} / \text{num_of_shots})$.

data cleaning/imputation (missing values, outliers, ...) 2/2

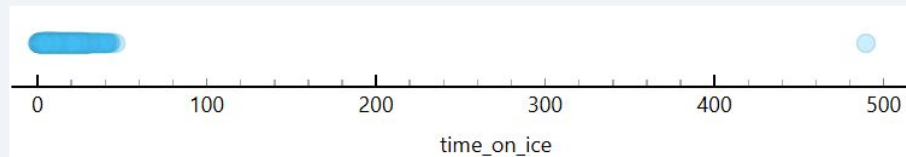
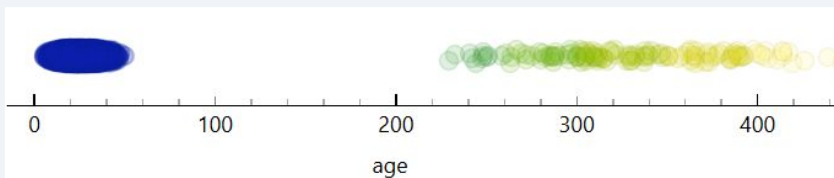
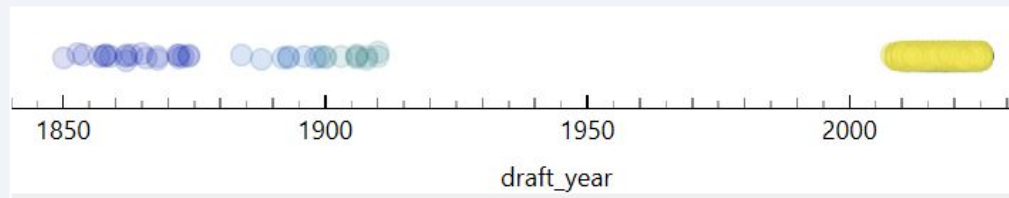
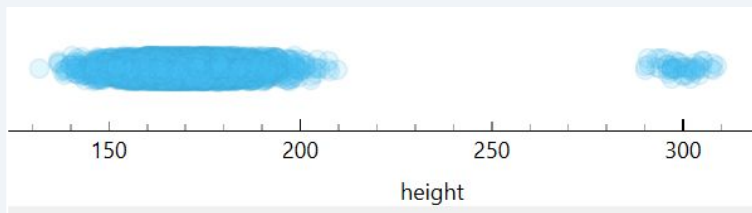
Outliers were assigned the value of NaN (not deleted; retains other valid information for each individual player).

Age greater than 200: 118 entries (maximum age of 455 reduced to 39)

Draft year less than 1950: 40 entries (minimum draft year of 1850 reduced to 2008)

Time on ice greater than 100: 2 entries (maximum of 490 reduced to 45.5)

Height > 250cm: 50 entries (maximum was 309cm - physically impossible)

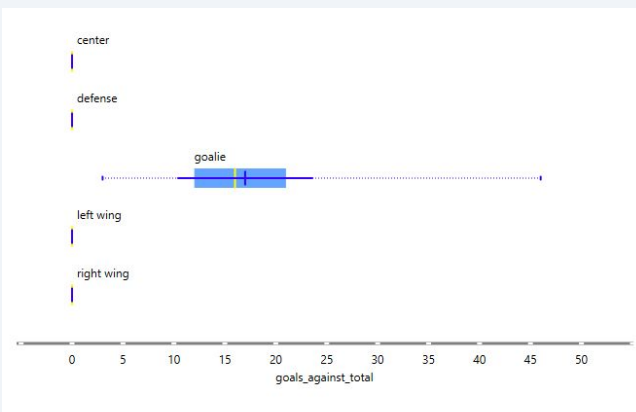


Note that the graphs were created with some jitter to better show dense clouds.

most important findings

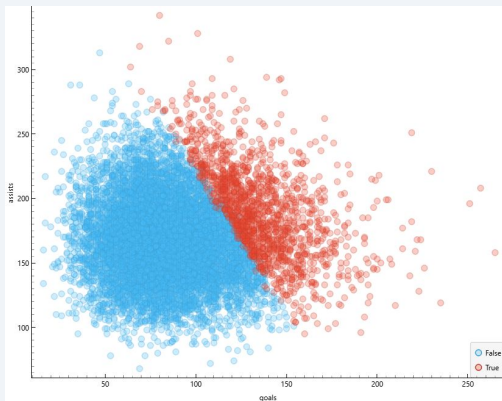
For some features, the position plays an essential role.

For example (graph), goals_against is 0 for all positions other than goalie. A trained model may consider this.

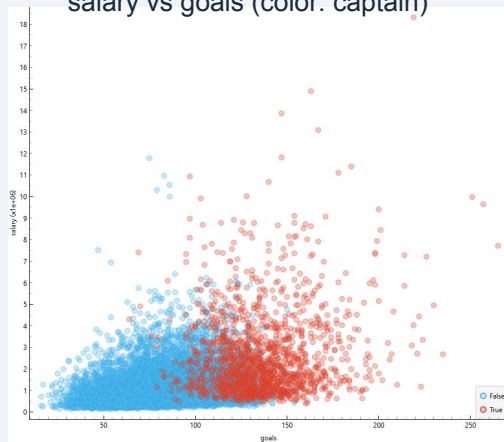


Furthermore, the feature **is_captain** is closely related to **assists**, **goals**, **salary** as seen in the graphs below. (red = captain)

assists vs goals (color: captain)



salary vs goals (color: captain)



There are some weird but not impossible values with num_of_shots < 300 and shooting_percentage_calculated > 65 (red).

